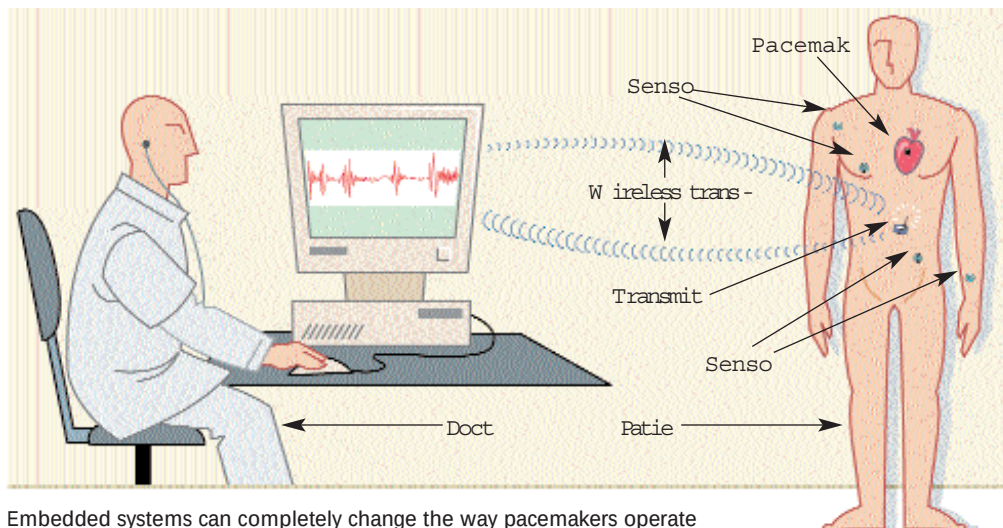


## What goes into an Embedded System?

An embedded system is any device controlled by instructions stored on a chip. These devices are usually controlled by a microprocessor that executes the instructions stored on a Read Only Memory (ROM) chip.

One of the most popular Real Time Operating Systems (RTOS) in use today is QNX (pronounced 'queue nicks'). It is used for everything from medical instrumentation and monitoring nuclear reactors, to traffic lights and industrial process control. In fact, it is so widely used that we use devices having QNX several times a day without being aware of it.

QNX makes use of a micro kernel as opposed to OSes such as Windows and UNIX, so system level functions such as device drivers are not part of the system. The kernel contains a minimum number of features for implementing basic system calls. These include message passing along with other interprocess communication, interrupt handling support, and other features. The other functions of the operating system are provided optionally and modularly through processes that communicate mostly through message passing. Message passing is a two-way asymmetric synchronised exchange of data between two processes in the OS. It is called asymmetric because one of the processes must be a 'sender' and the other must be a 'receiver' for passing to take place.



Embedded systems can completely change the way pacemakers operate

a mobile phone that can be worn as jewellery. The components of the phone will be distributed among different pieces of jewellery—earring, necklace, ring and bracelet.

The phone is likely to have Bluetooth capability built into it. The earrings will have embedded speakers and will act as the receiver. The necklace will have embedded microphones that will act as the mouthpiece users can talk into. IBM calls the ring part of the phone the 'decoder ring'. Light emitting diodes (LEDs) will flash to indicate an incoming call. The ring will also have features that will enable it to be programmed to flash different colours for a particular user or to indicate the importance of a call. A video graphics array (VGA) will be built into the bracelet, which will display the name and phone number of the caller. The bracelet will also integrate the keypad and dialling functions in it. IBM plans to incorporate voice recognition technology for dialling a number. The phone may also have features to indicate new e-mail.

### Don't keep your eyes on the road

Embedded systems can also make driver-less vehicle control a reality. Major automobile manufacturers are already engaged in work on these concepts. One such technology is Adaptive Cruise Control (ACC) from Ford.

ACC allows cars to keep safe distances from other vehicles on busy highways. The driver can set the speed of his car and the distance between his car and others. When traffic slows down, ACC alters vehicle speed using moderate braking. This ensures that a constant distance is maintained between cars. As soon as traffic becomes less, ACC moves up to the desired cruise speed that has been set by the driver. The driver can override the system anytime he wants to by braking.

Each car with ACC has a microwave radar

unit or laser transceiver fixed in front of it to determine the distance and relative speed of any vehicle in its path. The ACC computer (what else but an embedded system or a grouped system of embedded systems) constantly controls the throttle and brakes of the car. This helps to make sure that the set cruise speed or adapted speed of traffic at that time is not exceeded.

Another revolution is the way Internet services will be integrated into the car. So when you drive past your mechanic's, you will be reminded that your engine oil needs a refill, and when you cross the city limits, the toll will automatically get deducted from your bank account. And while passing the shopping mall, your PDA,



A mobile phone that you can wear as jewellery

which is connected to the Net via the car, will inform you about a new sale. In fact, the automatic toll deduction concept is already in effect in several countries around the globe.

All these application areas are just tiny drops in the big ocean of embedded systems technology. These proverbial 'Davids' are all set to conquer a world that is forbidden territory for the popular desktop OS 'Goliaths'. So wait with bated breath for the fireworks to come. They are sure to blow your mind. 